

Ours model output correct while baseline output wrong

Case 1:

Question: "Albert is wondering how much pizza he can eat in one day.
He buys 2 large pizzas and 2 small pizzas. A large pizza has 16
slices and a small pizza has 8 slices. If he eats it all, how
many pieces does he eat that day?",

Ground Truth:
"48"

[Planner Output (Generated by Qwen-AR)]

Summary: "Total 4 steps: Step 1 - calculate slices in 2 large pizzas
; Step 2 - calculate slices in 2 small pizzas; Step 3 - add
slices from large and small pizzas; Step 4 - finalize total
slices eaten.", "steps": ["Step 1: Multiply the number of slices
in one large pizza (16) by the number of large pizzas (2).", "
Step 2: Multiply the number of slices in one small pizza (8) by
the number of small pizzas (2).", "Step 3: Add the total slices
from large pizzas to the total slices from small pizzas.", "Step
4: Finalize the total number of slices Albert eats."]

[CBD Final Output]

Step 1: Multiply the number of slices in one large pizza (16) by the
number of large pizzas (2).\n\nEach large pizza has 16 slices,
and Albert buys 2 large pizzas. So, the total number of slices
from the large pizzas is $16 * 2 = 32$ \n\nAlbert will eat
32 slices from the large pizzas.\n\nStep 2: Multiply the number
of slices in one small pizza (8) by the number of small pizzas
(2).\n\nEach small pizza has 8 slices, and Albert buys 2 small
pizzas. So, the total number of slices from the small pizzas is
 $8 * 2 = 16$ \n\nAlbert will eat a total of 16 slices from
the small pizzas.\n\nStep 3: Add the total slices from large
pizzas to the total slices from small pizzas.\n\nTo find the
total number of slices Albert eats, we add the slices from the
large pizzas (32) to the slices from the small pizzas (16).\n $32 + 16 = 48$ \n\nTherefore, Albert eats a total of 48 slices that
day.\n\nStep 4: Finalize the total number of slices Albert eats.
\n\nThe total number of slices Albert eats is 48.\n\n### 48

[Baseline Output]

He buys 2 large pizzas because $2 * 1 = 2$ \nHe buys 4 small
pizzas because $2 * 2 = 4$ \nHe eats 32 pieces of cheese
because $2 * 16 = 32$ \nHe eats 32 pieces of cheese
because $4 * 8 = 32$ \nHe eats 64 pieces in total because
 $32 + 32 = 64$ \n\n### 64

Ours model output correct while baseline output wrong Case 2:

Question: "A tomato plant has 100 tomatoes. Jane picks 1/4 of that number for use in their house. After a week, she goes back and picks 20 more tomatoes, and the following week picks twice that number. What's the total number of fruits remaining on the tomato plant?",

Ground Truth:
"15"

[Planner Output (Generated by Qwen-AR)]

Summary: "Total 5 steps: Step 1 - calculate 1/4 of 100; Step 2 - subtract 1/4 of 100 from the total; Step 3 - subtract 20 tomatoes; Step 4 - subtract twice 20 tomatoes; Step 5 - determine the remaining number of tomatoes.", "steps": ["Step 1: Calculate 1/4 of 100 to find how many tomatoes Jane picks initially.", "Step 2: Subtract the number of tomatoes picked in Step 1 from the initial total of 100.", "Step 3: Subtract 20 tomatoes picked in the first week from the result in Step 2.", "Step 4: Subtract twice 20 tomatoes (40 tomatoes) picked in the second week from the result in Step 3.", "Step 5: Determine the number of tomatoes remaining on the plant after all subtractions ."]

[CBD Final Output]

Step 1: Calculate 1/4 of 100 to find how many tomatoes Jane picks initially.\n\nTo find 1/4 of 100, we divide 100 by 4:\n\n $100 \div 4 = 25$ \n\nSo, Jane picks 25 tomatoes initially.\n\n#### Jane picks 25 tomatoes from the tomato plant.\n\nStep 2: Subtract the number of tomatoes picked in Step 1 from the initial total of 100.\n\nAfter picking 25 tomatoes, we subtract 25 from the initial total of 100:\n\n $100 - 25 = 75$ \n\n#### After Jane's initial pick of 25 tomatoes, there are 75 tomatoes remaining on the plant.\n\nStep 3: Subtract 20 tomatoes picked in the first week from the result in Step 2.\n\nJane picks 20 more tomatoes in the first week, so we subtract 20 from the remaining 75 tomatoes:\n\n $75 - 20 = 55$ \n\n#### After Jane picks 20 tomatoes, there are 55 tomatoes remaining on the plant.\n\nStep 4: Subtract twice 20 tomatoes (40 tomatoes) picked in the second week from the result in Step 3.\n\nTwice 20 tomatoes is 40 tomatoes. We subtract 40 from the remaining 55 tomatoes:\n\n $55 - 40 = 15$ \n\n#### After Jane picks another 40 tomatoes, there are 15 tomatoes remaining on the plant.\n\nStep 5: Determine the number of tomatoes remaining on the plant after all subtractions.\n\nThe total number of tomatoes remaining on the tomato plant is 15.\n\n

15

[Baseline Output]

When she picks $\frac{1}{4}$ of the number of tomatoes, she goes home with
 $\frac{1}{4} \times 100 = \langle \langle \frac{1}{4} \times 100 = 25 \rangle \rangle 25$ tomatoes. \n The total number of
tomatoes remaining on the plant will be $100 - 25 = \langle \langle 100 - 25 = 75 \rangle \rangle 75$
\n The next week she picks 20 more tomatoes, a total of $20 + 20 =$
 $\langle \langle 20 + 20 = 40 \rangle \rangle 40$ tomatoes. \n The total number of tomatoes remaining
on the plant will be $75 - 40 = \langle \langle 75 - 40 = 35 \rangle \rangle 35$ \n The next week she
picks twice the number she picked the previous week, that is
 $20 \times 2 = \langle \langle 20 \times 2 = 40 \rangle \rangle 40$ tomatoes. \n The total number of tomatoes
remaining on the plant will be $35 - 40 = \langle \langle 35 - 40 = 5 \rangle \rangle 5$ \n
5

Ours model output correct while baseline output wrong Case 3:

Question: "The Dow Jones Industrial Average fell 2% today. The Dow ended the day at 8,722. What was the Dow in the morning before the markets opened?",

Ground Truth:
"8900"

[Planner Output (Generated by Qwen-AR)]

Summary: "Total 3 steps: Step 1 - Identify percentage decrease; Step 2 - Represent final value as a percentage of the original; Step 3 - Solve for the original value.", "steps": ["Step 1: Identify that the Dow fell by 2%, meaning the final value is 98% of the original value.", "Step 2: Represent the final value of 8,722 as 98% of the original value using the equation $8,722 = 0.98 * \text{original}$.", "Step 3: Solve for the original value by dividing 8,722 by 0.98."]

[CBD Final Output]

Step 1: Identify that the Dow fell by 2%, meaning the final value is 98% of the original value.\n\nThe Dow Jones Industrial Average fell by 2%, which means that the final value is 2% less than the original value. This can be represented as $100\% - 2\% = 98\%$. Therefore, the final value is 98% of the original value.\n\nStep 2: Represent the final value of 8,722 as 98% of the original value using the equation $8,722 = 0.98 * \text{original}$.\n\nTo find the original value, we can represent the final value of 8,722 as 98% of the original value. This can be done using the equation $8,722 = 0.98 * \text{original}$, where "original" represents the original value.\n\nStep 3: Solve for the original value by dividing 8,722 by 0.98.\n\nTo find the original value, we need to divide 8,722 by 0.98. This will give us the initial value of the Dow Jones Industrial Average before the fall.\n\n $8,722 / 0.98 = 8,900$ \n\nTherefore, the Dow Jones Industrial Average was 8,900 in the morning before the markets opened.\n\n#### 8,900

[Baseline Output]

The Dow fell 2% to 8722 so it fell $8722 * .02 = <<8722 * .02=174>>174$ points\nThe Dow was $8722 - 174 = <<8722 - 174=8548>>8548$ in the morning\n#### 8548

Ours model output correct while baseline output correct Case 1:

Question: "Madison takes her dog to the dog park. Counting Madison's dog, half the dogs have spots and 1/5 have pointy ears. If 15 dogs have spots, how many have pointy ears?",
Ground Truth: "6",

[Planner Output (Generated by Qwen-AR)]

Summary: "Total 4 steps: Step 1 - calculate total number of dogs; Step 2 - determine fraction of dogs with pointy ears; Step 3 - calculate number of dogs with pointy ears; Step 4 - confirm calculations.", "steps": ["Step 1: Use the information that half the dogs have spots and 15 dogs have spots to calculate the total number of dogs.", "Step 2: Recognize that 1/5 of the total number of dogs have pointy ears.", "Step 3: Multiply the total number of dogs by 1/5 to find the number of dogs with pointy ears.", "Step 4: Double-check the calculations to ensure the result is consistent with the problem."],

[CBD Final Output]

Step 1: Use the information that half the dogs have spots and 15 dogs have spots to calculate the total number of dogs.\n\nSince half the dogs have spots and 15 dogs have spots, we can find the total number of dogs by multiplying 15 by 2. \n\n $15 * 2 = 30$ \n\nSo, there are 30 dogs in total at the dog park.\n\nStep 2: Recognize that 1/5 of the total number of dogs have pointy ears.\n\nFrom the given information, we know that 1/5 of the dogs have pointy ears. Since 1/5 of the total number of dogs have pointy ears, we can multiply the total number of dogs by 1/5 to find the number of dogs with pointy ears.\n\nStep 3: Multiply the total number of dogs by 1/5 to find the number of dogs with pointy ears.\n\nTo find the number of dogs with pointy ears, we multiply the total number of dogs (30) by 1/5. This can be calculated as:\n\n $30 * 1/5 = 6$ \n\nSo, there are 6 dogs with pointy ears.\n\nStep 4: Double-check the calculations to ensure the result is consistent with the problem.\n\nTo double-check the calculations, we can verify that the number of dogs with pointy ears is consistent with the given information. Since 1/5 of the dogs have pointy ears, and there are 30 dogs in total, 1/5 of 30 is indeed 6. This confirms that our calculation is correct.\n\n### 6

[Baseline Output]

There are $15 * 2 = <<15*2=30>>30$ dogs at the dog park.\nThere are $30 / 5 = <<30/5=6>>6$ dogs with pointy ears.\n### 6

Ours model output correct while baseline output correct Case 2:

Question: "On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. How many baseball cards does he have on Thursday?",
Ground Truth: "32",

[Planner Output (Generated by Qwen-AR)]

Summary: "Total 4 steps: Step 1 - calculate cards after Tuesday; Step 2 - calculate cards after Wednesday; Step 3 - calculate cards bought on Thursday; Step 4 - calculate total cards on Thursday.", "steps": ["Step 1: Divide Buddy's Monday baseball cards by 2 to find how many he has left after losing half on Tuesday.", "Step 2: Add 12 baseball cards to Buddy's Tuesday total to find his Wednesday total.", "Step 3: Divide Buddy's Tuesday total by 3 to determine how many cards he buys on Thursday.", "Step 4: Add Buddy's Wednesday total to the number of cards bought on Thursday to find his Thursday total."],

[CBD Final Output]

Step 1: Divide Buddy's Monday baseball cards by 2 to find how many he has left after losing half on Tuesday.\n\nBuddy's Monday baseball cards = 30\n\nTo find the number of cards Buddy has after losing half on Tuesday, divide 30 by 2:\n\n $30 \div 2 = 15$ \n\nBuddy has 15 baseball cards left after losing half on Tuesday.\n\nStep 2: Add 12 baseball cards to Buddy's Tuesday total to find his Wednesday total.\n\nBuddy's Tuesday total = 15\n\nCards bought on Wednesday = 12\n\nTo find Buddy's Wednesday total, add 12 to his Tuesday's total:\n\n $15 + 12 = 27$ \n\nBuddy has 27 baseball cards on Wednesday.\n\nStep 3: Divide Buddy's Tuesday total by 3 to determine how many cards he buys on Thursday.\n\nBuddy's Tuesday total = 15\n\nTo find the number of cards Buddy buys on Thursday, divide 15 by 3 (since he buys a third of his Tuesday total):\n\n $15 \div 3 = 5$ \n\nBuddy buys 5 baseball cards on Thursday.\n\nStep 4: Add Buddy's Wednesday total to the number of cards bought on Thursday to find his Thursday total.\n\nBuddy's Wednesday total = 27\n\nCards bought on Thursday = 5\n\nTo find Buddy's Thursday total, add 27 to his Wednesday's total:\n\n $27 + 5 = 32$ \n\nBuddy has 32 baseball cards on Thursday.\n\n#### 32

[Baseline Output]

On Tuesday Buddy has $30 / 2 = <<30/2=15>>15$ baseball cards.\n\nOn Wednesday Buddy has $15 + 12 = <<15+12=27>>27$ baseball cards.\n\nOn Thursday Buddy buys $15 / 3 = <<15/3=5>>5$ baseball cards.\n\nOn Thursday Buddy has $27 + 5 = <<27+5=32>>32$ baseball cards.\n\n#### 32

Our model output wrong while baseline output wrong:

```
"question": "Julio makes a mocktail every evening. He uses 1
              tablespoon of lime juice and tops with 1 cup of sparkling water.
              He can usually squeeze 2 tablespoons of lime juice per lime.
              After 30 days, if limes are 3 for $1.00, how much will he have
              spent on limes?",
"answer_text": "5"
```

[Planner Output (Generated by Qwen-AR)]

Summary: "Total 6 steps: Step 1 - calculate total tablespoons of lime juice needed; Step 2 - calculate total limes needed; Step 3 - determine number of lime sets; Step 4 - calculate cost per set of limes; Step 5 - calculate total cost; Step 6 - finalize calculation.", "steps": ["Step 1: Multiply the number of days (30) by the lime juice used per day (1 tablespoon) to find the total tablespoons of lime juice needed.", "Step 2: Divide the total tablespoons of lime juice needed by the juice yield per lime (2 tablespoons per lime) to find the total number of limes needed.", "Step 3: Divide the total number of limes needed by the number of limes per set (3 limes per set) to find the number of lime sets required.", "Step 4: Determine the cost per set of limes, which is \$1.00 for 3 limes.", "Step 5: Multiply the number of lime sets by the cost per set to calculate the total cost.", "Step 6: Finalize the total cost calculation to ensure accuracy."],

[CBD Final Output]

Step 1: Multiply the number of days (30) by the lime juice used per day (1 tablespoon) to find the total tablespoons of lime juice needed.\n\nJulio uses 1 tablespoon of lime juice per day. Over 30 days, the total amount of lime juice needed is calculated as follows: $30 \times 1 = 30$ tablespoons.\n\nJulio needs 30 tablespoons of lime juice for 30 days.\n\nStep 2: Divide the total tablespoons of lime juice needed by the juice yield per lime (2 tablespoons per lime) to find the total number of limes needed.\n\nEach lime yields 2 tablespoons of juice. To find the number of limes needed, divide the total tablespoons by 2: $30 \div 2 = 15$ limes.\n\nJulio needs 15 limes to have enough lime juice for 30 days.\n\nStep 3: Divide the total number of limes needed by the number of limes per set (3 limes per set) to find the number of lime sets required.\n\nEach set contains 3 limes. To find the number of sets required, divide the total number of limes needed by the number of limes per set (3): $15 \div 3 = 5$ sets.\n\nJulio needs 5 sets of limes.\n\nStep 4: Determine the cost per set of limes, which is \$1.00 for 3 limes.\n\nThe cost for 3 limes is \$1.00. To find the cost per set, divide the total cost by 3: $1.00 \div 3 = 0.33$ (rounded to two decimal places). The cost per set of limes is \$0.33.\n\nStep

5: Multiply the number of lime sets by the cost per set to calculate the total cost.\n\nJulio needs 10 sets of limes. To find the total cost, multiply 10 by the cost per set: $10 \times \$1.00 = \10.00 .\n\nThe total cost for limes is \$10.00.\n\nStep 6: Finalize the total cost calculation to ensure accuracy.\n\nThe total cost calculation is accurate, as it matches the expected result.\n\n#### 10

[Baseline Output]

He uses 1 tablespoon of lime juice per drink and can get 2 tablespoons of lime juice per lime so he needs $1/2 = <<1/2=0.5>>0.5$ limes per drink\nHe uses 0.5 limes per drink and over 30 days he will use $0.5 \times 30 = <<0.5 \times 30=15>>15$ limes\n3 limes cost \$1.00 and he needs 15 limes so he will need $15/3 = <<15/3=5>>5$ limes\n3 limes cost \$1.00 and he needs 5 limes so he will spend $5/3 = \$<<5/3=1.67>>1.67$ on limes\n#### 1.67